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# Statistical Models
# Week 11: Using rjags for MCMC, diagnosing convergence

comments

> **!** Extended lectures on Wednesday
>
> Wednesday in this week 11 (November 12th), we will again have longer of lectures, that is from 14-17.
> This is to make up for the fact that there were no lectures on Wednesday, November 5th (course week 10).

In the Wednesday lecture, we introduce the *Metropolis-Hastings-algorithm* ("MH"). We start with the general principle and an example for its implementation. The algorithm has an interesting history; if time allows, we will also touch upon that. Then we discuss variants of MH-sampling: we will see that using a symmetric proposal density makes calculations a bit simpler. One can also combine Gibbs sampling with MH, which is important in Bayesian analysis for multivariate parameters, when the priors are not conjugate to the likelihood.

All these algorithms can be used "out of the box''; there are various software options. You will learn to use the software JAGS for Bayesian data analysis. To this end, you need to describe your model in a well defined way; I will discuss the general structure of JAGS model descriptions and we look at a few examples. You can download the JAGS user manual here and on brightspace.

Finally, we look at statistics that help to assess if the Markov chains have converged and are mixing well: *trace plots*, the *Geweke* and *Gelman statistics*. The *effective sample size* tells us how much information we get out of a long Markov chain as compared to independent sampling. This is usually less, due to autocorrelation. We will discuss points to consider if we want to improve convergence speed and decrease autocorrelation (R&G, Section 3.4.2, 3.4.3).

## Relevant Readings

- ☐ Metropolis-Hastings algorithm: R&G, Section 3.2.2,
- ☐ JAGS: R&G, Section 3.3
- ☐ Assessing and improving convergence speed: R&G, Section 3.4, in particular 3.4.2 and 3.4.3

## Install JAGS

From this week on, you will use the R package `rjags` for Bayesian analysis. `rjags` is an interface to the software JAGS (Just Another Gibbs Sampler) that you need to install separately on your computer (not from within R). Please install JAGS from sourceforge and feel free to ask for help if you experience problems with that.

## Lecture Material

- Link to onenotes on Google drive
- Coding the Metropolis Hastings algorithm

## Exercises

tba